Parametric Survival Analysis

Rats Dataset — Time to Tumour Onset

Outline

What we cover today:

1. Dataset & Study Design
2. Descriptive Summary
3. Kaplan–Meier Analysis
4. Cox Regression Models
5. Parametric Model Selection
6. Life Functions
7. Frailty Model
8. Conclusions

Key Research Question

Does a carcinogenic treatment affect time to tumour onset in rats, and which parametric model best describes this survival process?

Approach: Non-parametric exploration → Cox regression → Parametric modelling → Frailty

Dataset & Study Design

Source: `rats` dataset, `survival` package in R

Variables used:

- `time` — days to tumour or censoring
- `event` — 1 = tumour, 0 = censored
- `treatment` — Control vs Treated (rx)
- `sex` — female / male covariate
- `litter` — clustering variable (100 litters)

Design: 2:1 allocation
200 Control, 100 Treated
Max follow-up = 104 days

Treated rats received a carcinogenic agent (rx = 1)

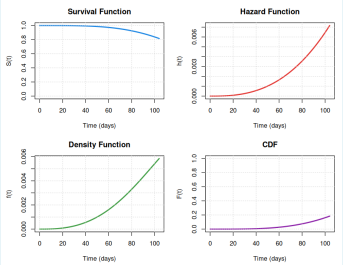
Sample Summary

Quantity	Value
Total rats	300
Events (tumours)	42
Censored	258
Overall event rate	14%
Control events	21/200 (10.5%)
Treated events	21/100 (21.0%)
Mean survival	90.4 days
Missing values	None

Note: Low event rate (14%) means median survival is not estimable — we report RMST instead.

Descriptive Summary

Survival Time Distribution



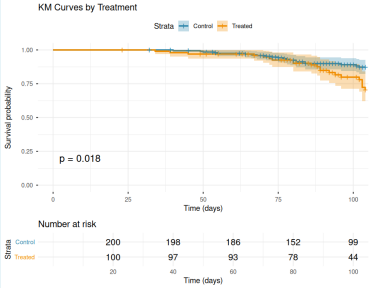
Key observations:

- Survival times are **left-skewed** — most rats survived near the 104-day limit
- Treated group has **double the event rate** (21% vs 10.5%)
- Sex is **highly imbalanced** in tumour rates: female rats develop far more tumours
- No missing values — clean dataset

Group	RMST (days)	SE
Overall	99.78	0.72
Control	100.38	0.82
Treated	98.55	1.40

Kaplan–Meier Analysis

KM Curves by Treatment Group



Results:

- **Log-rank p = 0.018** — significant difference between groups
- Median survival **not reached** in either group (event rate <50%)
- Control RMST ≈**1.8 days longer** than treated
- **Sex** also significant ($p < 0.0001$) — males far more resistant to tumours

Why no median?

Only 14% of rats developed tumours — the KM curve never drops below 0.5, so the median is undefined. We report RMST instead.

Cox Proportional Hazards Models

Hazard Ratios for Treatment

Model	HR	95% CI	p
Unadjusted	2.042	1.12–3.74	0.021
+ Sex adjusted	2.206	1.20–4.04	0.011
Clustered SE	2.206	1.24–3.92	**0.007**

Sex Effect (adjusted model)

Variable	HR	p
Sex (male)	0.047	<0.001

Key findings:

- Treated rats have $\approx$**2**$\times$ **the hazard** of tumour onset across all models
- Male rats have **HR = 0.047** — 95% lower hazard than females — sex is the strongest predictor
- Clustering by litter (robust SE): results remain consistent — no inflation of significance
- Tie-handling: Breslow, Efron, Exact all give **identical results**

PH test: treatment $p = 0.020$ — possible violation, motivating parametric models.

Parametric Model Selection

Model Comparison by AIC

Model	AIC	
Weibull	**578.21**	$\star$
Log-Logistic	578.33	
Log-Normal	578.87	
Gamma	578.37	
Gen. Gamma	580.13	
Gompertz	580.65	
Exponential	629.55	

$\star$ Best model (lowest AIC)

Winner: Weibull model

MLE Parameter Estimates:

- Shape $k = 3.667$ 95% CI: [2.99, 4.49]
- Scale $\lambda = 160.6$ 95% CI: [148, 174]

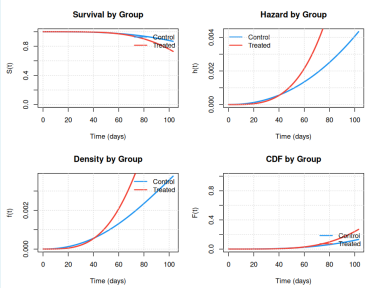
Shape Interpretation

$k > 1 \Rightarrow$ **Increasing hazard** — the longer a rat survives, the higher its risk of developing a tumour. Risk accelerates with time.

PP-plot confirmed good agreement between Weibull and empirical KM estimates.

Life Functions of the Weibull Model

2×2 Life Function Plots



Overall Weibull estimates:

- $E(T) = \lambda\,\Gamma(1 + 1/k) \approx$ **150.6 days**
- $\mathrm{Var}(T) \approx 2{,}180$ days^2
- $\mathrm{SD}(T) \approx 46.7$ days

By treatment group:

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Shape k	3.16	4.38
Scale λ	191.4	134.2
Mean (days)	173.2	124.8

Control rats remain tumour-free ≈**48 days longer** on average.

Frailty Model & PH Diagnostics

Why frailty?
Rats from the same litter share genetics and environment — outcomes are *not independent*. Standard Cox inflates Type I error.

Shared Gamma Frailty Results

- HR (Treated) = **2.069**, p = 0.022
- Frailty variance θ = **2.02**

⇒ *Substantial litter clustering*

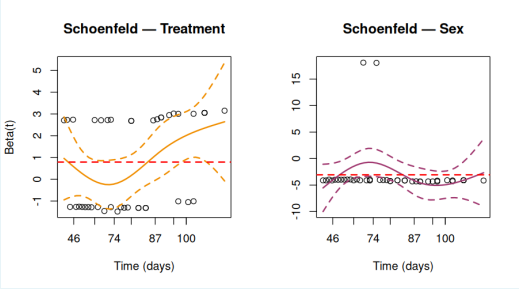
- Concordance = **0.923**

PH Assumption (Schoenfeld test)

- Treatment: p = 0.020 ⇒ violated
- Sex: p = 0.438 ⇒ holds

PH violation justifies the Weibull/AFT approach over standard Cox.

Schoenfeld Residuals



A non-flat pattern in the treatment residuals confirms a time-varying hazard ratio — the treatment effect changes over the study period.

Conclusions

What we found:

1. **Treatment doubles tumour risk** — HR $\approx$ 2.1–2.2 across all models ($p < 0.05$)
2. **Sex is the dominant predictor** — males 95% less likely to develop tumours (HR $= 0.047$, $p < 0.001$)
3. **Weibull fits best** (AIC $= 578.21$), shape $k = 3.667$ confirms increasing hazard
4. **Litter clustering is substantial** ($\theta = 2.02$) — frailty model essential for valid inference
5. **PH assumption violated** for treatment — parametric/AFT models preferred over standard Cox

Final Summary

Result	Value
Log-rank p	0.018
Cox HR (treated)	2.21
Best model	Weibull
Weibull AIC	578.21
Shape k	3.667
Mean (Control)	173.2 d
Mean (Treated)	124.8 d
Frailty θ	2.02
Frailty HR	2.069

Full code & analysis on GitHub:
walternyamutamba-svg/
rats-survival-analysis